

Higher-Rank Completion Holonomy: Displacement-Marked Optimal Fibers and Physical LIFO Transport

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Abstract

We develop a higher-rank transfer theory from optimal-fiber topology to explicitly specified physical pushdown transport. For a finite spanning lattice-convex optimal fiber $P \subset L$, the basic intrinsic object is the displacement-marked group $\mathfrak{M}(P) = (\pi_1(Q(P)), L, \delta_P)$, where δ_P records completion-lattice displacement. We prove affine-lattice invariance and exact sequences separating the full zero-displacement sector K_0 from its abelian shadow K_{lin} . Product, prism, and apex rank-raising admit exact marked-group laws: prism preserves both pure sectors, whereas apex preserves K_{lin} but creates new nonabelian pure holonomy. We also exhibit fixed-rank families with constant marked group and unbounded higher Betti numbers, showing that higher homology is not determined by pure displacement holonomy. After fixing a finite-permutation transport and a buffered literal-order carrier category, we identify the physically realized pure image, derive persistent carrier-girth bounds, an exact support-based ancestry law, a connection-dependent transient-state bound, and a general repair formulation as nearest-flat nonabelian cocycle optimization. Finally, minimal fibers in every active rank admit an S_3 -valued transport with nontrivial pure A_3 -holonomy, exact literal carrier height $s + 2$, and two ancestry handles. The resulting theory is layered rather than scalar: marked topology determines the available pure-loop domain, while physical representation and repair costs depend on the chosen transport category and connection.

Keywords. completion geometry; optimal fibers; displacement marking; holonomy; deterministic pushdown normalization; LIFO transport; periodic quotients; permutation monodromy.

1 Introduction

Shortest-completion problems for deterministic pushdown normalization naturally produce finite sets of optimal completion vectors. After translation, such a fiber may be viewed as a finite lattice-convex configuration $P \subset L$, where the active lattice L records equal-length completion trades. The periodic quotient $Q(P)$ captures the intrinsic identifications generated by those trades. In low rank, the topology of $Q(P)$ already distinguishes qualitatively different behaviors: a loop can represent genuine lattice displacement, or it can return to zero displacement while remaining topologically nontrivial. The latter phenomenon is the natural source of completion holonomy that is invisible to the semantic endpoint.

The purpose of this paper is to develop that distinction in arbitrary active rank and then connect it to explicitly specified physical pushdown transport. The central point is that the unmarked fundamental group is not the correct invariant. A loop in $Q(P)$ carries a canonical lattice displacement, and this marking must be retained. We therefore work with

$$\mathfrak{M}(P) = (\pi_1(Q(P)), L, \delta_P),$$

where

$$\delta_P : \pi_1(Q(P)) \twoheadrightarrow L$$

records the endpoint displacement of a lifted loop. Its kernel

$$K_0(P) = \ker \delta_P$$

is the full pure zero-displacement sector, while

$$K_{\text{lin}}(P) = \ker \delta_P^{\text{ab}}$$

is the part visible after abelianization. This separation is already substantial: $K_{\text{lin}} = 0$ does not imply $K_0 = 1$, and the same abstract fundamental group can occur with different displacement visibility.

The first half of the paper develops an exact higher-rank calculus for these marked data. Affine lattice equivalence preserves $\mathfrak{M}(P)$, and the pure sectors fit into the exact sequences

$$1 \longrightarrow [G, G] \longrightarrow K_0(P) \longrightarrow K_{\text{lin}}(P) \longrightarrow 0,$$

$$0 \longrightarrow K_{\text{lin}}(P) \longrightarrow H_1(Q(P); \mathbb{Z}) \longrightarrow L \longrightarrow 0.$$

Thus K_{lin} is precisely the abelian shadow of the full pure sector. We then compute the behavior of the marked group under the principal rank-raising operations. Products split coordinatewise. A prism adds a displacement-visible $\mathbb{Z}$ -direction and preserves both pure sectors. An apex adds a free generator: it leaves K_{lin} unchanged but creates new zero-displacement commutators. This gives a sharp intrinsic distinction between two operations that raise active rank by the same amount.

The marking is still only first-order topological information. We show that even the full marked group does not determine higher homology. For every fixed rank $s \geq 2$ we construct a family with

$$\mathfrak{M}(P_{m,s}) \cong (\mathbb{Z}^s, \mathbb{Z}^s, \text{id})$$

for all m , hence with trivial full and linear pure sectors, while every higher Betti number β_j , $2 \leq j \leq s$, grows without bound. This separates displacement holonomy from higher-dimensional topology and prevents higher Betti numbers from being used as a surrogate measure of pure-loop complexity.

The second half of the paper fixes physical categories instead of attributing memory cost to topology alone. For a finite permutation representation

$$\rho : \pi_1(Q(P)) \rightarrow S_n,$$

we define the physically realized pure image

$$\Pi_\rho(P) = \rho(K_0(P)).$$

The marked group determines which pure loops are available; the representation determines which of them remain physically visible. At this level the prism/apex distinction persists: prism preserves the realized pure image, whereas an apex generator sent to $h \in S_n$ creates exactly the normal closure generated by the old pure image and the commutators $[\rho(G), h]$.

To connect finite holonomy with stack resources, we then freeze a buffered literal-order carrier category. Persistent sheet identity is stored only in the order of a common zero-semantic Parikh vector. This yields an arithmetic carrier-girth lower bound. The support of the permutation image gives an exact ancestry-handle law, while transient path memory is controlled by a separate collision multiplicity of the chosen local connection. Finally, correction of a raw comparison policy is formulated as a nearest-flat nonabelian 1-cocycle problem on the 2-skeleton. These quantities therefore belong to different layers of the realization problem and should not be collapsed into a single scalar notion of “topological memory.”

Relation to existing frameworks. Several ingredients used below are classical in isolation, and our claims are not that loop monodromy, group-valued edge labels, or structured stack-height disciplines are new. In covering-space theory, actions of the fundamental group on a fiber and permutation representations $\pi_1(X) \rightarrow S_n$ are standard monodromy data [1]. Permutation voltage assignments and gain graphs likewise encode coverings and cycle products by group labels [2, 3]. In the theory of periodic nets, quotient cycles may carry translation or shift data, with zero-shift cycles forming a natural kernel of the translation bookkeeping [4, 5]. At a more algorithmic level, cycle consistency of group-valued relative measurements is central to group synchronization and related repair problems [6]. Finally, visibly pushdown and height-deterministic pushdown automata provide established models in which stack-height evolution is structurally constrained [7, 8].

Our use of these ideas is specific to completion-fiber quotients. We package the intrinsic data as the displacement-marked group $\mathfrak{M}(P)$ and derive exact product, prism, and apex laws inside the lattice-convex optimal-fiber class. The physical half then fixes a literal-order carrier category and separates pure-image, carrier, ancestry, transient-state, and repair coordinates. Thus the contribution is the completion-specific transfer calculus and its exact resource consequences, rather than the underlying notions of monodromy, gain, periodic shift, or cycle consistency themselves.

The main contributions can be summarized as follows.

- (i) For completion fibers, we package the displacement-marked group $\mathfrak{M}(P)$ as the intrinsic higher-rank datum and prove its affine-lattice invariance together with the universal visibility exact sequences.
- (ii) We establish exact product, prism, and apex transfer laws. Prism preserves the full pure sector, while apex preserves only its abelian shadow and can create new nonabelian zero-displacement commutators.
- (iii) We give fixed-rank families with constant marked group and unbounded higher homology, separating higher Betti data from displacement holonomy.
- (iv) After fixing finite-permutation transport, we determine the realized pure image and its prism/apex transfer laws.
- (v) In the buffered literal-order category we derive carrier-girth bounds, an exact support-based ancestry law, a connection-dependent transient-state bound, and a nearest-flat repair formulation.
- (vi) We construct an explicit family in every active rank with an S_3 -valued transport, nontrivial pure A_3 -holonomy, exact literal carrier height $s + 2$, and exact ancestry cost two.

The scope is deliberately category-sensitive. Nontrivial K_0 means that pure loops exist intrinsically; it does not force every physical representation to realize them. Likewise, carrier height, ancestry handles, transient control, and correction distance are not invariants of $Q(P)$ alone. The paper’s conclusion is therefore a layered transfer principle rather than a topology-to-memory identity:

$$\begin{aligned} \text{marked topology} &\longrightarrow \text{chosen holonomy representation} \longrightarrow \text{carrier/locality constraints} \\ &\longrightarrow \text{connection memory and repair.} \end{aligned}$$

Sections 2–5 develop the intrinsic marked theory; Sections 6–8 pass to the physical categories; and Section 9 gives the exact all-rank family.

2 Periodic Optimal Fibers and Displacement Marking

We begin at the intrinsic level. Let L be a free abelian group of finite rank and let $P \subset L$ be finite, spanning, and lattice-convex. Thus

$$P = \text{conv}(P) \cap L, \quad \langle P - P \rangle_{\mathbb{Z}} = L.$$

After translating P we may and shall assume $0 \in P$. The translation changes neither the periodic quotient below nor the marked displacement data, up to the affine-lattice identifications formalized in Section 3.

Definition 2.1 (Periodic optimal-fiber nerve). The *periodic nerve* $\mathcal{N}(P)$ is the simplicial complex with vertex set L in which a finite set $F \subset L$ is a simplex if and only if there exists $c \in L$ such that

$$c - F \subset P.$$

The lattice L acts freely on vertices by translation and simplicially on $\mathcal{N}(P)$. We write

$$Q(P) := |\mathcal{N}(P)|/L$$

for the corresponding periodic translation quotient.

The quotient is finite because P is finite. Its fundamental group admits a convenient presentation in terms of differences inside P . For $p \in P$ we use X_p for the oriented edge class of completion displacement p ; the precise sign convention is fixed in the proof below.

Proposition 2.2 (Difference presentation). *For $0 \in P$ there is a presentation*

$$\pi_1(Q(P)) \cong \langle X_p \ (p \in P) \mid X_0 = 1, \quad X_p^{-1}X_q = X_u^{-1}X_v \text{ whenever } q - p = v - u \rangle.$$

Proof. Every simplex of $\mathcal{N}(P)$ is contained in a translate $c - P$ of the full simplex on the vertex set $-P$. Translation by L identifies all quotient vertices, and cells of dimension at least three do not affect the fundamental group, so it is enough to read the edge identifications and triangle relations in the quotient 2-skeleton.

We use a completion-displacement sign convention. For $p \in P$, let X_p denote the quotient class of the oriented edge

$$c \longrightarrow c - p$$

in a translate $c - P$. Its completion displacement is the initial lattice point minus the terminal lattice point, namely p . The case $p = 0$ is degenerate, giving $X_0 = 1$.

For $p, q \in P$, the vertices $c, c - p, c - q$ form a 2-simplex. Its triangle relation expresses the oriented edge from $c - p$ to $c - q$ as $X_p^{-1}X_q$. The completion displacement of that edge is

$$(c - p) - (c - q) = q - p.$$

Two oriented edges in translates of the fundamental simplex are identified in the periodic quotient exactly when their translations, equivalently their completion displacements, agree. Hence the edge represented by $X_p^{-1}X_q$ is identified with the edge represented by $X_u^{-1}X_v$ precisely when

$$q - p = v - u.$$

Conversely, every quotient edge identification arises in this way, and the triangle relations have already been used to express all edge classes in the displayed generators. The presentation follows. $\square$

The repeated-difference relations encode the local translation identifications in the periodic nerve. We shall use this presentation as the algebraic interface between the quotient topology and lattice displacement.

Definition 2.3 (Displacement marking). Let

$$G_P := \pi_1(Q(P)).$$

The *displacement homomorphism* is the map

$$\delta_P : G_P \longrightarrow L, \quad \delta_P(X_p) = p.$$

Equivalently, for a lifted quotient loop we record initial lattice point minus terminal lattice point, matching the completion-displacement convention above.

Proposition 2.4. *The map δ_P is a well-defined surjective homomorphism.*

Proof. A defining relation

$$X_p^{-1}X_q = X_u^{-1}X_v$$

is imposed only when $q - p = v - u$. Hence both sides have the same image under the proposed map to L , so δ_P is well defined. Since $0 \in P$ and P spans L , the subgroup of L generated by the elements $p \in P$ is all of L . The images of the generators therefore generate L , proving surjectivity. $\square$

The displacement marking separates two logically different phenomena. A loop may be nontrivial in $Q(P)$ because it translates in the active lattice, or it may be nontrivial while returning to the same lattice point. The latter sector is the intrinsic source of pure completion holonomy.

Definition 2.5 (Pure-loop sectors). Define

$$K_0(P) := \ker \delta_P$$

and, after abelianization,

$$K_{\text{lin}}(P) := \ker \left(H_1(Q(P); \mathbb{Z}) \xrightarrow{\delta_P^{\text{ab}}} L \right).$$

We call $K_0(P)$ the *full pure zero-displacement sector* and $K_{\text{lin}}(P)$ its *linear* or *abelian visibility sector*.

Thus $K_0(P)$ remembers every zero-displacement loop, including nonabelian ones that disappear after abelianization, whereas $K_{\text{lin}}(P)$ records only those pure directions that remain visible in first homology. The distinction will be essential: higher-rank families can have $K_{\text{lin}} = 0$ while retaining substantial nonabelian pure holonomy, and conversely large higher homology need not produce any pure holonomy at all.

Remark 2.6 (Scope). The objects in this section are intrinsic to the periodic completion quotient. They do not, by themselves, impose a memory cost on an arbitrary pushdown implementation. Physical carrier, ancestry, transient-state, and repair costs will be introduced only after an explicit transport category is fixed.

2.1 Realization as genuine optimal fibers

For completeness we record the bridge back to integer completion systems. If $A \in \mathbb{Z}^{k \times n}$ and $r \in \mathbb{Z}^k$, write

$$\mathcal{O}_A(r) := \operatorname{argmin}\{\mathbf{1}^T x : x \in \mathbb{N}^n, Ax = -r\}$$

whenever the feasible set is nonempty. After choosing $x_0 \in \mathcal{O}_A(r)$, the normalized optimal fiber is the finite configuration

$$P_A(r) := \mathcal{O}_A(r) - x_0$$

in its difference lattice.

Proposition 2.7 (Lattice-convex realization). *Every finite lattice-convex configuration $P \subset L$ spanning a rank- s lattice is, up to affine lattice isomorphism, the normalized optimal fiber of an integer counting system. The realizing matrix can be chosen with a strictly positive integer kernel vector, so every integer fiber in its image has a nonnegative representative after adding a large multiple of that vector.*

Proof. Identify L with $\mathbb{Z}^s$ and translate so that $0 \in P$. Choose integral affine facet functions

$$\ell_j(z) = b_j - a_j^T z \quad (1 \leq j \leq m)$$

for the lattice polytope $\text{conv}(P)$, so that

$$\text{conv}(P) = \{z \in \mathbb{R}^s : \ell_j(z) \geq 0 \ \forall j\}.$$

Choose $B \in \mathbb{Z}^s$ large enough that $z + B \in \mathbb{N}^s$ for every $z \in P$ and set

$$E_0(z) := (z + B, \ell_1(z), \dots, \ell_m(z)) \in \mathbb{Z}^{s+m}.$$

Because $P = \text{conv}(P) \cap \mathbb{Z}^s$, one has

$$E_0(z) \geq 0 \iff z \in P \quad (z \in \mathbb{Z}^s).$$

Choose

$$C \geq \max_{z \in P} \mathbf{1}^T E_0(z)$$

and append a balancing coordinate,

$$E(z) := (E_0(z), C - \mathbf{1}^T E_0(z)) \in \mathbb{Z}^n, \quad n = s + m + 1.$$

Then $E(z) \geq 0$ exactly for $z \in P$, and $\mathbf{1}^T E(z) = C$ for every lattice point for which $E(z)$ is nonnegative.

Let

$$D := \{E(z) - E(0) : z \in \mathbb{Z}^s\} \subset \mathbb{Z}^n.$$

Projection to the first s coordinates is a left inverse, so $D \cong \mathbb{Z}^s$ is a saturated direct summand; moreover every element of D has coordinate sum zero. Append one final coordinate and put

$$h = (1, \dots, 1) \in \mathbb{Z}^{n+1}, \quad K := (D \times \{0\}) + \mathbb{Z}h.$$

The last coordinate of h is one, so K is saturated. Hence there is an integer matrix A with $\ker_{\mathbb{Z}} A = K$.

Set $x_0 = (E(0), 0)$ and $r = -Ax_0$. Every integer point in the affine fiber of x_0 has a unique form

$$x = x_0 + (d, 0) + th, \quad d \in D, \quad t \in \mathbb{Z}.$$

Nonnegativity forces $t \geq 0$ from the last coordinate, while

$$\mathbf{1}^T x = C + t(n+1).$$

Thus every minimum-length point has $t = 0$. Writing $d = E(z) - E(0)$, nonnegativity at $t = 0$ is equivalent to $z \in P$. Therefore the minimum fiber is precisely

$$\mathcal{O}_A(r) = \{(E(z), 0) : z \in P\},$$

and $z \mapsto (E(z), 0)$ is an affine lattice isomorphism onto its normalization. The strictly positive vector $h \in \ker_{\mathbb{Z}} A$ provides the stated completion-totality property. $\square$

Thus the lattice-convex examples used below are not merely abstract quotient complexes: they occur as genuine normalized optimal completion fibers.

3 The Displacement-Marked Group

The unmarked group $\pi_1(Q(P))$ does not retain enough information for higher-rank transport questions: the same abstract group may occur with different lattice displacement maps. The correct intrinsic object is therefore the group together with its displacement marking.

Definition 3.1 (Displacement-marked group). For a finite spanning lattice-convex configuration $P \subset L$, define

$$\mathfrak{M}(P) := (G_P, L, \delta_P), \quad G_P = \pi_1(Q(P)), \quad \delta_P : G_P \twoheadrightarrow L.$$

A marked isomorphism

$$(G, L, \delta) \cong (G', L', \delta')$$

means a pair of isomorphisms (ϕ, A) with

$$A \circ \delta = \delta' \circ \phi.$$

3.1 Affine-lattice invariance

Theorem 3.2 (Affine-lattice invariance). *Let $f : P \rightarrow P'$ be an affine lattice equivalence*

$$f(z) = Az + b,$$

where $A : L \xrightarrow{\sim} L'$ is a lattice isomorphism and $P' = A(P) + b$. Then there is a quotient homeomorphism

$$\bar{A} : Q(P) \xrightarrow{\cong} Q(P')$$

whose induced map on fundamental groups satisfies

$$\boxed{\delta_{P'} \circ (\bar{A})_* = A \circ \delta_P.}$$

Consequently $\mathfrak{M}(P)$ is an affine-lattice invariant up to marked isomorphism.

Proof. Define a vertex map on the periodic nerves by

$$\tilde{A} : L \rightarrow L', \quad z \mapsto Az.$$

Suppose $F \subset L$ is a simplex of $\mathcal{N}(P)$. Then some $c \in L$ satisfies $c - F \subset P$, and therefore

$$(Ac + b) - AF = A(c - F) + b \subset A(P) + b = P'.$$

Thus AF is a simplex of $\mathcal{N}(P')$. Applying the inverse affine lattice isomorphism gives the inverse simplicial map. Moreover

$$\tilde{A}(z + \ell) = \tilde{A}(z) + A\ell,$$

so the map is equivariant for the translation actions and descends to a homeomorphism of quotients.

Lift a based loop in $Q(P)$ to a path in $|\mathcal{N}(P)|$. If its endpoint differs from its initial point by $\ell \in L$, then the lifted image path in $|\mathcal{N}(P')|$ has endpoint displacement $A\ell$. Hence

$$\delta_{P'}((\bar{A})_*[\gamma]) = A\delta_P([\gamma]),$$

which is the desired marked compatibility. □

3.2 Visibility exact sequences

The marking gives a universal algebraic decomposition of pure holonomy into its nonabelian and abelian parts.

Theorem 3.3 (Universal visibility exact sequences). *Let*

$$G := G_P = \pi_1(Q(P)), \quad \delta := \delta_P : G \twoheadrightarrow L.$$

Then

$$1 \longrightarrow [G, G] \longrightarrow K_0(P) \longrightarrow K_{\text{lin}}(P) \longrightarrow 0$$

is exact. Moreover

$$0 \longrightarrow K_{\text{lin}}(P) \longrightarrow H_1(Q(P); \mathbb{Z}) \xrightarrow{\delta^{\text{ab}}} L \longrightarrow 0$$

is exact.

Proof. Because the target L is abelian, every commutator lies in the kernel of δ ; hence

$$[G, G] \subseteq K_0(P).$$

Let

$$\alpha : G \rightarrow G_{\text{ab}} = H_1(Q(P); \mathbb{Z})$$

be abelianization. Restriction gives a homomorphism

$$\alpha|_{K_0} : K_0(P) \rightarrow K_{\text{lin}}(P).$$

It is surjective: if $[g] \in K_{\text{lin}}(P)$, then $\delta^{\text{ab}}([g]) = 0$, equivalently $\delta(g) = 0$, so $g \in K_0(P)$. Its kernel is

$$K_0(P) \cap [G, G] = [G, G],$$

since $[G, G] \subseteq K_0(P)$. This proves the first exact sequence.

The second is immediate from the definition of $K_{\text{lin}}(P)$ together with surjectivity of δ_P , which implies surjectivity of δ^{ab} because L is already abelian. $\square$

Corollary 3.4. *For every P as above:*

(i)

$$K_0(P)/[G_P, G_P] \cong K_{\text{lin}}(P).$$

(ii) *If $K_{\text{lin}}(P) = 0$, then*

$$K_0(P) = [G_P, G_P].$$

(iii) *$K_0(P) = 1$ if and only if G_P is abelian and $K_{\text{lin}}(P) = 0$.*

Proof. Part (i) is the first isomorphism theorem applied to $\alpha|_{K_0}$. Part (ii) follows immediately. For (iii), if $K_0 = 1$ then $[G_P, G_P] \subseteq K_0$ forces G_P to be abelian and of course $K_{\text{lin}} = 0$. Conversely, if G_P is abelian then $[G_P, G_P] = 1$, and if also $K_{\text{lin}} = 0$, part (ii) gives $K_0 = 1$. $\square$

Since L is free abelian, the second exact sequence in Theorem 3.3 splits as a sequence of abelian groups, although no canonical splitting is selected by the intrinsic data. Hence

$$H_1(Q(P); \mathbb{Z}) \cong K_{\text{lin}}(P) \oplus L$$

noncanonically. In particular every torsion class of $H_1(Q(P); \mathbb{Z})$ lies in $K_{\text{lin}}(P)$.

Definition 3.5 (Linear visibility excess). Define

$$\nu(P) := \text{rank } K_{\text{lin}}(P).$$

Then

$$\boxed{\nu(P) = \beta_1(Q(P)) - \text{rank } L.}$$

The invariant $\nu(P)$ measures the rank of the linearly visible pure sector, but it does not measure the full nonabelian pure sector. In particular, $\nu(P) = 0$ is compatible both with $K_0(P) = 1$ and with a large commutator subgroup.

4 Higher-Rank Marked-Group Calculus

We now show that the standard rank-raising operations admit exact laws at the level of displacement-marked groups. Once $\mathfrak{M}(P)$ is known, these operations can be computed without returning to the underlying counting realization.

4.1 Products

The periodic nerve itself admits a product law, not merely its fundamental group.

Proposition 4.1 (Topological product law). *Let $P \subset L$ and $R \subset M$ be finite spanning lattice-convex configurations. Then $P \times R$ is lattice-convex in $L \oplus M$, spans a lattice of rank $\text{rank } L + \text{rank } M$, and*

$$\boxed{Q(P \times R) \simeq Q(P) \times Q(R).}$$

The homotopy equivalence is induced by an $(L \oplus M)$ -equivariant equivalence before passing to the quotient.

Proof. Write $K = \mathcal{N}(P)$ and $H = \mathcal{N}(R)$. A finite set $F \subset L \times M$ is a simplex of $\mathcal{N}(P \times R)$ exactly when some $(c, d) \in L \times M$ satisfies

$$(c, d) - F \subset P \times R.$$

This is equivalent to

$$c - \pi_L(F) \subset P, \quad d - \pi_M(F) \subset R.$$

Thus $\mathcal{N}(P \times R)$ is the simplicial categorical product determined by the two coordinate projections.

On the posets of nonempty faces define

$$\phi(F) = (\pi_L F, \pi_M F), \quad \psi(A, B) = A \times B.$$

Then $\phi\psi = \text{id}$ and

$$F \subseteq \psi\phi(F)$$

for every face F . After barycentric subdivision, pointwise comparable poset maps induce homotopic simplicial maps [9]; hence ϕ and ψ give mutually inverse homotopy equivalences. The order complex of the product face poset is the standard barycentric triangulation of the product of the two barycentric subdivisions, so the realization is naturally homeomorphic to $|K| \times |H|$. All constructions commute with translations, which gives the asserted equivariance and therefore the quotient equivalence.

Finally

$$\text{conv}(P \times R) = \text{conv}(P) \times \text{conv}(R),$$

so intersection with $L \oplus M$ gives exactly $P \times R$, proving lattice convexity. The difference lattice is the direct sum of the two factor difference lattices, which gives the rank statement. $\square$

For marked groups define

$$(G, L, \delta) \boxtimes (H, M, \varepsilon) := (G \times H, L \oplus M, \delta \oplus \varepsilon).$$

Theorem 4.2 (Product transfer). *Let $P \subset L$ and $R \subset M$ be finite spanning lattice-convex configurations. Then*

$$\mathfrak{M}(P \times R) \cong \mathfrak{M}(P) \boxtimes \mathfrak{M}(R).$$

Equivalently, there is a canonical isomorphism

$$\Phi_{P,R} : \pi_1(Q(P \times R)) \xrightarrow{\sim} \pi_1(Q(P)) \times \pi_1(Q(R))$$

such that

$$\delta_{P \times R} = (\delta_P \oplus \delta_R) \circ \Phi_{P,R}.$$

Consequently

$$K_0(P \times R) \cong K_0(P) \times K_0(R),$$

$$K_{\text{lin}}(P \times R) \cong K_{\text{lin}}(P) \oplus K_{\text{lin}}(R),$$

and

$$\nu(P \times R) = \nu(P) + \nu(R).$$

Proof. Translate so that $0 \in P$ and $0 \in R$. In the difference presentation of $G_{P \times R}$ put

$$A_p := X_{(p,0)}, \quad B_r := X_{(0,r)}.$$

The equality of differences

$$(p, r) - (p, 0) = (0, r) - (0, 0)$$

gives

$$A_p^{-1} X_{(p,r)} = B_r,$$

so

$$X_{(p,r)} = A_p B_r.$$

Likewise

$$(p, 0) - (0, 0) = (p, r) - (0, r)$$

gives

$$A_p = B_r^{-1} X_{(p,r)} = B_r^{-1} A_p B_r,$$

so A_p commutes with B_r for every p, r .

The elements A_p satisfy precisely the repeated-difference relations for P , and the B_r satisfy those for R . Define

$$\Phi_{P,R}(X_{(p,r)}) = (X_p, X_r).$$

This is well defined because equality of differences in $L \oplus M$ is equivalent to equality of differences in each coordinate. Conversely, the inclusions

$$X_p \mapsto A_p, \quad X_r \mapsto B_r$$

define homomorphisms $G_P \rightarrow G_{P \times R}$ and $G_R \rightarrow G_{P \times R}$ with commuting images, hence a homomorphism

$$\Psi : G_P \times G_R \rightarrow G_{P \times R}.$$

The identities $X_{(p,r)} = A_p B_r$ and the definitions of A_p, B_r show $\Psi \Phi_{P,R} = \text{id}$ and $\Phi_{P,R} \Psi = \text{id}$. Thus $\Phi_{P,R}$ is an isomorphism.

For a generator,

$$(\delta_P \oplus \delta_R) \Phi_{P,R}(X_{(p,r)}) = (p, r) = \delta_{P \times R}(X_{(p,r)}),$$

so the marked compatibility follows. Taking kernels gives the K_0 formula; taking abelianizations gives the K_{lin} formula. The identity for ν follows by ranks. $\square$

4.2 Prism rank-raising

Let

$$I := \{0, 1\} \subset \mathbb{Z}.$$

Its difference presentation has one nontrivial generator and no nontrivial relation, so

$$\mathfrak{M}(I) \cong (\mathbb{Z}, \mathbb{Z}, \text{id}).$$

Corollary 4.3 (Prism preservation). *For every finite spanning lattice-convex P ,*

$$K_0(P \times I) \cong K_0(P), \quad K_{\text{lin}}(P \times I) \cong K_{\text{lin}}(P).$$

Moreover

$$\nu(P \times I) = \nu(P).$$

Thus prism raises active rank by one while adding only a displacement-visible direction. It creates no new pure zero-displacement sector.

Proof. Apply Theorem 4.2 and use $K_0(I) = 1$, $K_{\text{lin}}(I) = 0$. □

4.3 Apex rank-raising

Define

$$P^+ := (P \times \{0\}) \cup \{e\} \subset L \oplus \mathbb{Z}e, \quad e = (0, 1).$$

The apex operation also raises active rank by one, but its marked-group behavior is fundamentally different from that of the prism.

Theorem 4.4 (Apex pure-holonomy creation). *Let $G = G_P = \pi_1(Q(P))$. Then P^+ is lattice-convex, spans $L \oplus \mathbb{Z}e$, and has active rank $\text{rank } L + 1$. Moreover*

$$\pi_1(Q(P^+)) \cong G * \langle t \rangle,$$

and under this identification the displacement map is

$$\delta_+(g) = (\delta_P(g), 0) \quad (g \in G), \quad \delta_+(t) = (0, 1).$$

Furthermore

$$K_0(P^+) = \langle\langle K_0(P), [G, t] \rangle\rangle_{G * \langle t \rangle},$$

while

$$K_{\text{lin}}(P^+) \cong K_{\text{lin}}(P), \quad \nu(P^+) = \nu(P).$$

If $\text{rank } L > 0$, then P^+ contains genuinely new nonabelian pure holonomy not contained in the embedded old kernel $K_0(P)$.

Proof. Translate the base so that $0 \in P$. Every lattice point of $\text{conv}(P^+)$ has last coordinate either 0 or 1. The height-zero slice has lattice points exactly $P \times \{0\}$, while the height-one slice consists only of the apex e . Hence P^+ is lattice-convex. Its difference lattice contains $L \times \{0\}$ together with e , so it is $L \oplus \mathbb{Z}e$.

In the difference presentation for P^+ , write

$$A_p := X_{(p,0)} \quad (p \in P), \quad t := X_e.$$

The relations among the A_p are exactly those defining G . No nontrivial repeated-difference relation involves t . Indeed, base–base differences have last coordinate 0, apex–base differences have last coordinate +1, and base–apex differences have last coordinate −1. Equality of two +1

differences forces their base points to agree, and the same is true for two -1 differences. Thus every apex-involving relation is tautological, proving

$$\pi_1(Q(P^+)) \cong G * \langle t \rangle.$$

The formula for δ_+ follows directly from the generators.

Put $K = K_0(P)$ and

$$N := \langle\langle K, [G, t] \rangle\rangle.$$

Clearly $N \subseteq \ker \delta_+$, since K has zero base displacement and the target $L \oplus \mathbb{Z}$ is abelian. Conversely, quotienting first by the normal closure of K gives

$$(G * \langle t \rangle) / \langle\langle K \rangle\rangle \cong L * \langle t \rangle,$$

because $G/K \cong L$. Quotienting further by all cross-commutators $[L, t]$ gives

$$L \times \langle t \rangle \cong L \oplus \mathbb{Z}.$$

The induced map to the displacement target is the identity on this group, so no further kernel remains. Hence $\ker \delta_+ = N$.

After abelianization,

$$H_1(Q(P^+); \mathbb{Z}) \cong H_1(Q(P); \mathbb{Z}) \oplus \mathbb{Z}[t],$$

and the induced displacement map is $\delta_P^{\text{ab}} \oplus \text{id}_{\mathbb{Z}}$. Therefore $K_{\text{lin}}(P^+) \cong K_{\text{lin}}(P)$ and $\nu(P^+) = \nu(P)$.

Finally assume rank $L > 0$. Choose $g \in G$ with $\delta_P(g) \neq 0$. Then $g \neq 1$, and the free-product normal form implies

$$[g, t] = g^{-1}t^{-1}gt \neq 1.$$

This commutator does not lie in the embedded factor G , hence is not in the old kernel $K_0(P) \subset G$, while $\delta_+([g, t]) = 0$. Thus apex creates a genuinely new nonabelian pure element. $\square$

Corollary 4.5 (Apex–prism separation). *For the same nonzero-rank base P , prism and apex both raise active rank by one and both preserve K_{lin} and ν , but*

$$K_0(P \times I) \cong K_0(P),$$

whereas

$$K_0(P^+) = \langle\langle K_0(P), [G_P, t] \rangle\rangle$$

contains new nonabelian pure elements. Thus prism is pure-sector preserving and apex is pure-sector creating.

4.4 Higher tori and iteration

Corollary 4.6 (Higher-torus zero-pure-holonomy). *For*

$$I^s = \{0, 1\}^s$$

we have

$$\boxed{Q(I^s) \simeq T^s}$$

and

$$\boxed{\mathfrak{M}(I^s) \cong (\mathbb{Z}^s, \mathbb{Z}^s, \text{id}),}$$

so

$$\boxed{K_0(I^s) = 1, \quad K_{\text{lin}}(I^s) = 0.}$$

In particular, arbitrarily high active rank and nonzero torus homology through the top degree do not by themselves create a pure zero-displacement sector.

Proof. The periodic nerve for $I = \{0, 1\}$ is the line with edges between consecutive integers, so $Q(I) \cong S^1$. Iterate Proposition 4.1 for the topological statement and Theorem 4.2 for the marked-group statement. $\square$

Proposition 4.7 (Iterated apex). *After m apex extensions of P ,*

$$G^{(m)} \cong G_P * F_m$$

with new free generators $t_1, \dots, t_m$, and the displacement target is $L \oplus \mathbb{Z}^m$. Moreover

$$K_0^{(m)} = \langle\langle K_0(P), [G_P, t_i] \ (1 \leq i \leq m), [t_i, t_j] \ (i \neq j) \rangle\rangle,$$

while

$$K_{\text{lin}}^{(m)} \cong K_{\text{lin}}(P), \quad \nu^{(m)} = \nu(P).$$

Proof. Apply Theorem 4.4 inductively. Each new generator contributes an independent coordinate in the displacement target, while every commutator among old and new free factors has zero displacement. Abelianization adds a free $\mathbb{Z}$ -summand mapped isomorphically to the new lattice coordinate at each step, so the linear kernel is unchanged. $\square$

Together, Theorem 4.2, Corollary 4.3, and Theorem 4.4 form the higher-rank displacement-kernel calculus. Products add marked data coordinatewise, prisms add a displacement-visible abelian direction without new pure holonomy, and apexes add a free direction that preserves the linear pure sector while creating nonabelian pure commutators.

5 Separation Phenomena

The marked-group calculus separates several kinds of information that are easily conflated at the level of homotopy type alone. We first show that the abstract fundamental group does not determine displacement visibility. We then exhibit families in which the entire marked group remains fixed while higher homology grows without bound. These examples locate precisely what is, and what is not, encoded by $\mathfrak{M}(P)$.

5.1 The fundamental group alone does not determine displacement visibility

Proposition 5.1 (Same fundamental group, different displacement visibility). *There exist finite spanning lattice-convex configurations $P_3 \subset \mathbb{Z}^3$ and $P_F \subset \mathbb{Z}^2$ such that*

$$\pi_1(Q(P_3)) \cong F_3 \cong \pi_1(Q(P_F)),$$

but

$$K_{\text{lin}}(P_3) = 0, \quad K_{\text{lin}}(P_F) \cong \mathbb{Z}.$$

Consequently the abstract group $\pi_1(Q(P))$ determines neither the active lattice rank nor the linear zero-displacement sector.

Proof. Take the rank-three minimal configuration

$$P_3 = \{0, e_1, e_2, e_3\} \subset \mathbb{Z}^3.$$

The four points are affinely independent, so no two distinct nonzero pair differences agree. Hence the repeated-difference presentation has no nontrivial relation among the three nonzero generators, and

$$\pi_1(Q(P_3)) \cong F_3.$$

The displacement map sends these free generators to the standard basis of $\mathbb{Z}^3$. After abelianization it is therefore an isomorphism

$$H_1(Q(P_3); \mathbb{Z}) \cong \mathbb{Z}^3 \xrightarrow{\sim} \mathbb{Z}^3,$$

so $K_{\text{lin}}(P_3) = 0$.

For the rank-two comparison, let

$$P_F = \{(0, 0), (0, 1), (1, 1), (3, 2)\} \subset \mathbb{Z}^2.$$

Its convex hull is the triangle with vertices $(0, 0)$, $(0, 1)$, $(3, 2)$, and $(1, 1)$ is its unique interior lattice point; hence P_F is lattice-convex. The six unordered nonzero pair differences are

$$(0, 1), (1, 1), (3, 2), (1, 0), (3, 1), (2, 1),$$

which are distinct up to sign. Thus the difference presentation again has no nontrivial relation among the three nonzero generators, giving

$$\pi_1(Q(P_F)) \cong F_3.$$

After abelianization the displacement map is a surjection

$$\mathbb{Z}^3 \twoheadrightarrow \mathbb{Z}^2.$$

Its kernel is therefore a primitive rank-one subgroup, so

$$K_{\text{lin}}(P_F) \cong \mathbb{Z}.$$

The two configurations have the same abstract fundamental group but different marked displacement data. $\square$

Remark 5.2. The proposition is already enough to rule out any formulation of higher-rank completion holonomy based on π_1 alone. The target lattice and the map to it are essential parts of the intrinsic datum.

5.2 A rank-two seed with growing second homology

We next construct a family whose marked group is fixed but whose second homology grows. For $m \geq 2$, define

$$P_m = \{(t, 0) : 0 \leq t \leq m\} \cup \{(0, 1), (m-1, -1)\} \subset \mathbb{Z}^2.$$

Write

$$H_m := \{(t, 0) : 0 \leq t \leq m\}$$

and let $A_m \subset Q(P_m)$ be the horizontal subcomplex represented by translates of faces contained in H_m .

Lemma 5.3 (Direct homological reduction for the seed). *For every $m \geq 2$,*

$$A_m \simeq S^1, \quad H_j(Q(P_m); \mathbb{Z}) = 0 \quad (j \geq 3),$$

and $H_2(Q(P_m); \mathbb{Z})$ is free abelian.

Proof. First consider the periodic nerve of the rank-one interval H_m . Its vertices may be identified with $\mathbb{Z}$, and a finite set $F \subset \mathbb{Z}$ is a simplex exactly when

$$\max F - \min F \leq m.$$

This is the nerve of the open interval cover

$$U_n = (n - \frac{1}{4}, n + m + \frac{1}{4}), \quad n \in \mathbb{Z},$$

of $\mathbb{R}$: because all endpoints are indexed by integers, a finite intersection is nonempty exactly under the displayed diameter condition. Every nonempty finite intersection is an interval, so the nerve lemma [9] gives a contractible realization. The horizontal translation subgroup $\mathbb{Z}$ acts freely, and its quotient is precisely A_m . Hence A_m is a $K(\mathbb{Z}, 1)$ and therefore

$$A_m \simeq S^1.$$

It remains to control the cells outside the horizontal subcomplex. Let Δ_m be the abstract simplex on the vertex set P_m , and let $B_m \subset \Delta_m$ be its horizontal face on H_m . Cells of $Q(P_m)$ are translation orbits of faces of Δ_m . We claim that every nonhorizontal face of dimension at least two has a unique representative in P_m . Indeed, suppose $S \subset P_m$ is nonhorizontal, $|S| \geq 3$, and $S + a \subset P_m$ for some $a = (a_x, a_y) \in \mathbb{Z}^2$. If $a_y = 0$, then S contains one of the unique off-line points at height 1 or -1 ; preservation of that height forces that point to be fixed, hence $a_x = 0$. If $a_y = 1$, then S contains no height-1 point, and at most one horizontal point can occur because all horizontal points would be sent to the unique height-1 point; together with the unique height- -1 point this gives $|S| \leq 2$, a contradiction. The case $a_y = -1$ is symmetric, while $|a_y| \geq 2$ is even more restrictive. Thus $a = 0$.

Let

$$\bar{C}_* = C_*(\Delta_m, B_m; \mathbb{Z}), \quad D_* = C_*(Q(P_m), A_m; \mathbb{Z}).$$

By the uniqueness just proved, for every $k \geq 2$ the groups D_k and $\bar{C}_k$ are canonically identified through the nonhorizontal k -faces. For $k \geq 3$ their boundary maps agree, because every codimension-one face then has dimension at least two (horizontal faces vanish in both relative complexes). Since both Δ_m and B_m are contractible and nonempty,

$$H_*(\Delta_m, B_m; \mathbb{Z}) = 0.$$

It follows immediately from the identical chain-complex tail that

$$H_k(Q(P_m), A_m; \mathbb{Z}) = 0 \quad (k \geq 3).$$

For degree two, only the edge identifications can change the boundary map. Thus there is a quotient homomorphism

$$q : \bar{C}_1 \longrightarrow D_1$$

with

$$\partial_2^D = q \partial_2^{\bar{C}}, \quad \partial_3^D = \partial_3^{\bar{C}}.$$

Because $H_2(\bar{C}_*) = 0$, one has

$$\ker \partial_2^{\bar{C}} = \text{im } \partial_3^{\bar{C}}.$$

Consequently

$$[z] \longmapsto \partial_2^{\bar{C}} z$$

defines an injection

$$H_2(Q(P_m), A_m; \mathbb{Z}) \hookrightarrow \bar{C}_1.$$

The target is free abelian, so the relative group is free abelian as well.

Finally use the long exact sequence of the pair $(Q(P_m), A_m)$. Since $A_m \simeq S^1$, its homology vanishes in degrees at least two. Together with the relative vanishing above this gives

$$H_j(Q(P_m); \mathbb{Z}) = 0 \quad (j \geq 3),$$

while

$$0 = H_2(A_m) \longrightarrow H_2(Q(P_m)) \longrightarrow H_2(Q(P_m), A_m)$$

is injective. Hence $H_2(Q(P_m); \mathbb{Z})$ is a subgroup of a free abelian group and is therefore free abelian. $\square$

Lemma 5.4 (Rank-two seed family). *For every $m \geq 2$, the configuration P_m is finite, spanning, and lattice-convex, and*

$$\mathfrak{M}(P_m) \cong (\mathbb{Z}^2, \mathbb{Z}^2, \text{id}).$$

Moreover

$$H_2(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^{m-1},$$

there is no homology above degree two, and the Betti polynomial is

$$1 + 2t + (m-1)t^2.$$

Proof. The convex hull of P_m is the lattice quadrilateral

$$(0, 0), (m-1, -1), (m, 0), (0, 1).$$

All four boundary edges are primitive and the Euclidean area is m . Pick's theorem therefore gives $m-1$ interior lattice points; they are exactly

$$(1, 0), \dots, (m-1, 0),$$

so the lattice points of the quadrilateral are precisely P_m . Thus P_m is lattice-convex and clearly spans $\mathbb{Z}^2$.

Write

$$x = X_{(1,0)}, \quad y = X_{(0,1)}, \quad z = X_{(m-1,-1)}.$$

Repeated horizontal differences give

$$X_{(t,0)} = x^t.$$

The repeated difference $(0, 1)$ between $((0, 0), (0, 1))$ and $((m-1, -1), (m-1, 0))$ gives

$$y = z^{-1}x^{m-1}, \quad z = x^{m-1}y^{-1}.$$

For $1 \leq t \leq m-1$, comparison of the two occurrences of the difference $(-t, 1)$ gives

$$x^{-t}y = yx^{-t}.$$

Taking $t = 1$ yields $xy = yx$. Hence the presentation is generated by the commuting pair x, y . Their displacements are the standard basis vectors of $\mathbb{Z}^2$, so no further relation is possible. Consequently

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2, \quad \delta_{P_m} = \text{id}$$

under these identifications. Thus $K_0(P_m) = 1$ and $K_{\text{lin}}(P_m) = 0$.

By Lemma 5.3, there is no homology above degree two and H_2 is free abelian. It remains only to compute its rank. The Euler characteristic can be computed from simplex orbits. The horizontal subcomplex has Euler characteristic zero. For either fixed off-line point, faces containing that point together with a nonempty horizontal subset contribute -1 , while faces containing both

off-line points have total alternating contribution zero. The only cross-identifications between the two one-off-line families are the m edge pairs

$$\{(0, 1), (t, 0)\} \sim \{(m-1, -1), (m-1-t, 0)\}, \quad 0 \leq t \leq m-1.$$

No higher-dimensional cross-identification occurs by the uniqueness argument in Lemma 5.3. Identifying the duplicate edge orbits raises the alternating count by m , and hence

$$\chi(Q(P_m)) = m - 2.$$

Since $\pi_1(Q(P_m)) \cong \mathbb{Z}^2$, one has $H_1 \cong \mathbb{Z}^2$, so $\beta_0 = 1$ and $\beta_1 = 2$. The vanishing above degree two gives

$$m - 2 = 1 - 2 + \beta_2,$$

whence $\beta_2 = m - 1$. Since H_2 is free,

$$H_2(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^{m-1}.$$

The Betti polynomial follows. □

5.3 Fixed marked group with unbounded higher homology

Theorem 5.5 (Fixed marked group with unbounded higher homology). *Fix $s \geq 2$ and set*

$$P_{m,s} := P_m \times I^{s-2}, \quad I = \{0, 1\}.$$

Then for every $m \geq 2$,

$$\boxed{\mathfrak{M}(P_{m,s}) \cong (\mathbb{Z}^s, \mathbb{Z}^s, \text{id}).}$$

In particular

$$K_0(P_{m,s}) = 1, \quad K_{\text{lin}}(P_{m,s}) = 0.$$

Nevertheless every Betti number

$$\beta_j(Q(P_{m,s})), \quad 2 \leq j \leq s,$$

is unbounded as $m \rightarrow \infty$. More precisely, the Betti polynomial is

$$\boxed{(1 + 2t + (m-1)t^2)(1+t)^{s-2}},$$

so

$$\beta_j = \binom{s-2}{j} + 2\binom{s-2}{j-1} + (m-1)\binom{s-2}{j-2},$$

with out-of-range binomial coefficients interpreted as zero.

Proof. Lemma 5.4 gives

$$\mathfrak{M}(P_m) \cong (\mathbb{Z}^2, \mathbb{Z}^2, \text{id}),$$

while Corollary 4.6 gives

$$\mathfrak{M}(I^{s-2}) \cong (\mathbb{Z}^{s-2}, \mathbb{Z}^{s-2}, \text{id}).$$

Applying the product theorem yields

$$\mathfrak{M}(P_{m,s}) \cong (\mathbb{Z}^2 \times \mathbb{Z}^{s-2}, \mathbb{Z}^2 \oplus \mathbb{Z}^{s-2}, \text{id}) \cong (\mathbb{Z}^s, \mathbb{Z}^s, \text{id}).$$

Thus the marked group is independent of m .

Topologically,

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}.$$

The homology of both factors is torsion free, so the integral Kunneth formula [1] multiplies Betti polynomials. Lemma 5.4 contributes $1 + 2t + (m - 1)t^2$, while T^{s-2} contributes $(1 + t)^{s-2}$. This gives the displayed formula. For each $2 \leq j \leq s$, the coefficient

$$\binom{s-2}{j-2}$$

of $m - 1$ is positive, so β_j is unbounded as $m \rightarrow \infty$. $\square$

Corollary 5.6 (Unbounded higher topology with trivial pure sector). *For every fixed active rank $s \geq 2$ there are configurations with arbitrarily large higher Betti numbers but*

$$K_0 = 1, \quad K_{\text{lin}} = 0.$$

Hence neither the presence nor the magnitude of higher homology supplies a positive lower bound on pure displacement-holonomy complexity.

Proof. Apply Theorem 5.5. $\square$

5.4 Exact determination hierarchy

Theorem 5.7 (Determination hierarchy). *Within the class of finite spanning lattice-convex completion fibers, the following separations hold.*

(i) *The abstract group $\pi_1(Q(P))$ does not determine active rank or $K_{\text{lin}}(P)$.*

(ii) *The marked group $\mathfrak{M}(P) = (\pi_1(Q(P)), L, \delta_P)$ determines*

$$K_0(P) = \ker \delta_P, \quad K_{\text{lin}}(P) = \ker \delta_P^{\text{ab}}$$

exactly.

(iii) *The linear sector $K_{\text{lin}}(P)$ does not determine the full pure sector $K_0(P)$.*

(iv) *Even the full marked group $\mathfrak{M}(P)$ does not determine the higher homology of $Q(P)$.*

Proof. Part (i) is Proposition 5.1. Part (ii) is tautological from the definition of the marking and Theorem 3.3. For part (iii), compare the higher torus I^s , where $K_{\text{lin}} = 0$ and $K_0 = 1$, with the minimal rank- s free configuration for $s \geq 2$, where $\pi_1 \cong F_s$, $K_{\text{lin}} = 0$, and Corollary 3.4 gives

$$K_0 = [F_s, F_s] \neq 1.$$

Part (iv) is Theorem 5.5. $\square$

The hierarchy shows that displacement marking is exactly the additional first-order datum needed to recover pure-loop visibility, but not a substitute for higher homology. Physical carrier, ancestry, transient-state, and repair coordinates are not part of the intrinsic datum at all; they are defined only after additional representation and connection data are fixed in the next sections. This is the point at which the intrinsic part of the theory ends and the category-dependent transport layer begins.

6 Finite Physical Holonomy

The preceding sections are intrinsic: they determine which zero-displacement loops exist, but not whether a chosen pushdown representation realizes those loops nontrivially. We now freeze a finite physical transport category and make this second layer explicit.

6.1 Marked finite-permutation transport

Definition 6.1 (Marked finite-permutation transport). Let

$$G = \pi_1(Q(P)), \quad \delta : G \rightrightarrows L,$$

and choose a permutation representation

$$\rho : G \longrightarrow S_n.$$

Define

$$G_{\text{phys}} := L \times S_n, \quad c : G_{\text{phys}} \rightarrow L, \quad c(\ell, \sigma) = \ell,$$

and

$$\text{Hol}_\rho : G \longrightarrow G_{\text{phys}}, \quad \text{Hol}_\rho(g) := (\delta(g), \rho(g)).$$

Then $c \circ \text{Hol}_\rho = \delta$. The *physically realized pure image* is

$$\Pi_\rho(P) := \rho(K_0(P)) \leq S_n.$$

The representation ρ is part of the physical model. Thus $K_0(P) \neq 1$ says that nontrivial pure loops are intrinsically available, not that every physical transport must see them. A representation may kill some or all of $K_0(P)$.

Proposition 6.2 (Flat connection realization). *Let X be a connected finite CW complex and let*

$$\rho : \pi_1(X, x_0) \rightarrow H$$

be a homomorphism to a finite group. Then the oriented edges of a finite CW subdivision of X admit an H -labeling whose product around every 2-cell is the identity and whose based loop holonomy is exactly ρ .

Proof. Choose a maximal tree T in the 1-skeleton and label every oriented tree edge by the identity, following the standard covering/monodromy construction [1]. For each oriented edge e outside T , join its endpoints to x_0 along the unique tree paths and let γ_e be the resulting based loop. Label e by $\rho([\gamma_e])$ and the reverse edge by its inverse. The standard tree collapse identifies the non-tree edges with generators of the fundamental group. The attaching word of every 2-cell represents the identity in $\pi_1(X, x_0)$, so applying ρ makes the ordered product of its edge labels equal to the identity. Conversely, the holonomy of a based loop is the image under ρ of its fundamental-group class. $\square$

Thus every abstract finite-permutation transport used below can be represented by a flat local edge connection; the homomorphism and the local labeling are two presentations of the same transport data.

Theorem 6.3 (Physical pure-image theorem). *For every marked finite-permutation transport,*

$$\boxed{\text{Hol}_\rho(K_0(P)) = \{0\} \times \Pi_\rho(P).}$$

Consequently the intrinsic marked group determines the available pure-loop domain, while ρ determines the part that survives physically:

$$\boxed{(G, L, \delta) \implies K_0(P) \xrightarrow{\rho} \Pi_\rho(P).}$$

Proof. If $k \in K_0(P)$ then $\delta(k) = 0$, so

$$\text{Hol}_\rho(k) = (0, \rho(k)).$$

As k ranges over $K_0(P)$, the second coordinate ranges exactly over $\rho(K_0(P)) = \Pi_\rho(P)$. $\square$

Remark 6.4 (Gauge dependence). Changing the physical trivialization objectwise conjugates the permutation representation. Hence the subgroup $\Pi_\rho(P)$ is defined only up to conjugacy, but its trivial or nontrivial status, its order, and the cardinality of its permutation support are invariant.

6.2 Prism and apex at the physical-image level

Theorem 6.5 (Physical prism preservation). *Let*

$$P^\square = P \times I, \quad I = \{0, 1\},$$

so that

$$\pi_1(Q(P^\square)) \cong G \times \langle u \rangle.$$

Suppose $\rho^\square : G \times \langle u \rangle \rightarrow S_n$ extends ρ , and write

$$h := \rho^\square(u).$$

Then h centralizes $\rho(G)$ and

$$\boxed{\Pi_{\rho^\square}(P^\square) = \Pi_\rho(P).}$$

Thus a prism may add displacement-visible monodromy, but it cannot create a new realized pure sector.

Proof. Because the domain is a direct product, u commutes with every $g \in G$, so $h = \rho^\square(u)$ centralizes $\rho(G)$. By Corollary 4.3,

$$K_0(P^\square) = K_0(P) \times 1.$$

Restricting $\rho^\square$ to this subgroup gives exactly $\rho|_{K_0(P)}$, and hence

$$\rho^\square(K_0(P^\square)) = \rho(K_0(P)).$$

□

Theorem 6.6 (Physical apex transfer). *Let P^+ be the apex extension of P . Choose $h \in S_n$ and extend ρ uniquely to*

$$\rho_h : G * \langle t \rangle \longrightarrow S_n, \quad \rho_h|_G = \rho, \quad \rho_h(t) = h.$$

Put

$$H_h := \langle \rho(G), h \rangle.$$

Then

$$\boxed{\Pi_{\rho_h}(P^+) = \langle\langle \Pi_\rho(P), [\rho(G), h] \rangle\rangle_{H_h} .}$$

Proof. Theorem 4.4 gives

$$K_0(P^+) = \langle\langle K_0(P), [G, t] \rangle\rangle_{G * \langle t \rangle} .$$

The homomorphism ρ_h maps the ambient free product onto H_h . Therefore the image of this normal closure is the normal closure in H_h of the images of the displayed generators. Those images are precisely

$$\rho(K_0(P)) = \Pi_\rho(P)$$

and the commutators $[\rho(g), h]$ for $g \in G$. This is the asserted formula. □

Corollary 6.7 (Centralizer criterion). *Under the hypotheses of Theorem 6.6:*

(i) *If $h \in C_{S_n}(\rho(G))$, then*

$$\Pi_{\rho_h}(P^+) = \Pi_\rho(P).$$

(ii) *If $\Pi_\rho(P) = 1$ and h does not centralize $\rho(G)$, then*

$$\Pi_{\rho_h}(P^+) \neq 1.$$

Thus the intrinsic apex supplies new zero-displacement commutators, while their physical visibility is controlled exactly by the chosen permutation image.

7 Literal Carrier and Locality Costs

We next pass from finite holonomy to a concrete persistent stack representation. The results in this section are relative to a buffered literal-order category; they are not unrestricted pushdown lower bounds.

7.1 Buffered literal-order carriers

Definition 7.1 (Buffered literal-order carrier). Fix an integer counting realization with matrix A whose semantic fiber contains P . A persistent carrier sheet is a literal stack word with Parikh vector $z \in \mathbb{N}^d$ satisfying

$$Az = 0.$$

Distinct persistent sheets have a common control state, common endpoint potential, and the same Parikh vector. Permanent sheet identity is stored only in literal stack order. Comparison macros are one-top and ancestry-bijective. During a macro, finite control may use temporary handles storing distinguished ancestry tokens; a one-top microstep may store the visible ancestry in an empty handle, restore a stored ancestry to the rewritten visible cell, exchange the visible ancestry with a stored handle, or push a new decorated cell carrying a stored ancestry. All handles are empty again at macro endpoints. Internal microsteps are allowed.

For $z \in \mathbb{N}^d$, define its order capacity by

$$M(z) := \frac{|z|!}{\prod_i z_i!}.$$

For $k \geq 2$, define the *literal order carrier girth*

$$g_k^{\text{ord}}(A) := \min\{|z| : z \neq 0, Az = 0, M(z) \geq k\},$$

with value $+\infty$ when the set is empty.

Theorem 7.2 (Persistent literal carrier bound). *Suppose a buffered literal-order implementation has k distinct persistent sheets with a common zero-semantic Parikh vector. If the persistent stack height is H , then*

$$H \geq g_k^{\text{ord}}(A).$$

In particular, in the regular-sheet realization of a finite permutation image $H_\rho := \text{im } \rho$, with one persistent sheet for each element of H_ρ ,

$$H \geq g_{|\text{im } \rho|}^{\text{ord}}(A).$$

Proof. If all persistent sheets have the same Parikh vector z , the number of distinct literal orderings available to encode sheet identity is at most

$$M(z) = \frac{|z|!}{\prod_i z_i!}.$$

Thus $k \leq M(z)$. Since $Az = 0$ and $z \neq 0$, the definition of $g_k^{\text{ord}}(A)$ gives $|z| \geq g_k^{\text{ord}}(A)$. The stack height of such a sheet is $H = |z|$, proving the first claim. The regular-sheet statement is the special case $k = |\text{im } \rho|$. $\square$

The bound depends on the chosen counting realization through A and on the chosen physical image through k . Active lattice rank enters only indirectly through those data.

7.2 Exact ancestry-handle law

For a finite permutation group $H \leq S_n$, define

$$\text{supp}(H) := \bigcup_{\sigma \in H} \text{supp}(\sigma), \quad a(H) := |\text{supp}(H)|.$$

Theorem 7.3 (Exact buffered ancestry law). *In the buffered ancestry-bijective macro category, the minimum number of temporary ancestry handles needed to realize every permutation in H on a contiguous active block is*

$$b_{\min}(H) = \max\{a(H) - 1, 0\}.$$

In particular, for every nontrivial H ,

$$b_{\min}(H) = a(H) - 1.$$

For a marked transport ρ this gives

$$b_{\min}(\rho) = \max\{a(\text{im } \rho) - 1, 0\}.$$

Proof. The trivial group has $a(H) = 0$ and requires no handle, so assume H is nontrivial. Then $a(H) \geq 2$. Place the $a(H)$ active distinguished cells contiguously at the top of the distinguished carrier block. By definition of the support, some $\sigma \in H$ moves the deepest active cell. Before a one-top ancestry-bijective macro can inspect that cell, the $a(H) - 1$ cells above it must have been removed from the live stack. Their ancestry classes must remain available for the final bijection, so at least $a(H) - 1$ temporary handles are necessary.

Conversely, $a(H) - 1$ handles suffice for every $\sigma \in H$: remove the upper $a(H) - 1$ active ancestries into the handles, expose the deepest active cell, and rebuild the active block in the target order, reattaching the stored ancestries as required. Inactive cells are untouched. Thus the lower bound is attained. $\square$

The support law is independent of the abstract order of the image group. Two representations of isomorphic groups may therefore have different ancestry costs if their permutation supports differ.

8 Connection Memory and Repair

Persistent carrier order need not encode all information needed during a local comparison macro. Path-dependent information can be transient, and a raw comparison policy may also fail the flatness constraints of the quotient. These two costs live at a layer below the marked group and the finite holonomy image.

8.1 Deferred path-memory multiplicity

Definition 8.1 (Terminal collision multiplicity). Let $\mathcal{C}$ be a deferred-carrier connection certificate: the persistent carrier order is not changed until the terminal update of the relevant macro. Define $\chi(\mathcal{C})$ to be the maximum size of a family of histories that

- (i) reach the same preterminal semantic vertex;
- (ii) expose the same persistent carrier sheet and the same next input or action; and
- (iii) have pairwise distinct accumulated physical multipliers requiring pairwise distinct terminal endpoint updates.

Theorem 8.2 (Deferred path-memory lower bound). *Let q_{tr} denote the number of transient path-memory classes available to such a deferred implementation. Then*

$$q_{\text{tr}} \geq \chi(\mathcal{C}).$$

If at every relevant semantic vertex the number of accumulated-product classes is at most $\chi(\mathcal{C})$, then $\chi(\mathcal{C})$ transient classes suffice, and the bound is exact.

Proof. Take a collision family of size $\chi(\mathcal{C})$. Immediately before the terminal update, all histories have the same semantic vertex, the same visible persistent carrier sheet, and the same next action. A deterministic transition rule supplied with the same physical configuration and the same transient class must make the same terminal update. Since the histories require pairwise distinct updates, they must occupy pairwise distinct transient classes. Hence $q_{\text{tr}} \geq \chi(\mathcal{C})$.

Conversely, if no relevant vertex supports more than $\chi(\mathcal{C})$ accumulated-product classes, assign one transient class to each local accumulated-product class and update it along the connection certificate. The semantic vertex and persistent sheet are already part of the surrounding configuration, so these labels suffice to distinguish every terminal action. Thus $\chi(\mathcal{C})$ classes are enough. $\square$

The quantity $\chi(\mathcal{C})$ depends on the chosen local connection and on the deferred update convention. It is not determined by (G, L, δ) or by ρ alone.

8.2 Repair as a nearest-flat nonabelian cocycle problem

Let X be a finite cell model for the relevant shortest-prefix quotient, let H be a finite physical comparison group, and choose one orientation from each unoriented edge; denote this set by $E^+(X)$. A raw comparison policy is an oriented-edge labeling

$$a : E^{\text{or}}(X) \longrightarrow H, \quad a(\bar{e}) = a(e)^{-1}.$$

Definition 8.3 (Flat labeling). A labeling b is *flat* if the ordered product of its labels around the attaching loop of every 2-cell is the identity. In a fixed trivialization, denote the set of flat labelings by

$$Z^1(X; H).$$

The notation is suggestive of nonabelian 1-cocycles; no abelian group structure on $Z^1(X; H)$ is assumed.

Definition 8.4 (Repair coordinates). Fix a conjugation-invariant generating metric d_H on H . For a raw labeling a , set

$$C_0(a) := \min_{b \in Z^1(X; H)} |\{e \in E^+(X) : a(e) \neq b(e)\}|,$$

$$C_1(a) := \min_{b \in Z^1(X; H)} \sum_{e \in E^+(X)} d_H(a(e), b(e)).$$

Additional coordinates, such as maximum support or maximum local correction radius, may be minimized simultaneously to obtain a repair Pareto set.

Proposition 8.5 (Two-skeleton suffices for flatness). *The group-valued flatness constraints are completely determined by the 2-skeleton $X^{(2)}$. Cells of dimension at least three impose no additional relations on the holonomy representation of $\pi_1(X)$.*

Proof. The fundamental group of a CW complex is determined by its 2-skeleton: the 1-skeleton provides generators and the attaching maps of the 2-cells provide relations, while attaching cells of dimension at least three does not change π_1 . A group-valued edge labeling is flat exactly when the product around each 2-cell attaching loop is the identity. Thus every group-valued flatness equation already occurs in $X^{(2)}$. $\square$

Proposition 8.6 (Repair is representation and presentation relative). *The values $C_0(a)$, $C_1(a)$, and their support/locality refinements are not determined by the intrinsic marked group $\mathfrak{M}(P)$ alone. They depend on the raw labeling a , the chosen physical group H , the fixed trivialization and cell presentation, and the metric d_H .*

Proof. The feasible set $Z^1(X; H)$ depends on the chosen physical group and cell presentation, while the objective functions explicitly use a and d_H . None of these data is part of $\mathfrak{M}(P)$. Changing them can alter either the feasible flat labelings or their distances from the raw policy without changing the intrinsic marked group. $\square$

Sections 6–8 therefore produce a layered rather than scalar transfer law:

marked topology $\longrightarrow$ chosen holonomy representation $\longrightarrow$ carrier/support constraints $\longrightarrow$ connection memory $\longrightarrow$ raw-policy repair cost
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The next section shows that this hierarchy admits an explicit all-rank family with nontrivial pure finite holonomy and exact carrier/locality costs.

9 An All-Rank Exact Family

The preceding sections separate intrinsic marked topology from the successive physical choices that turn it into a concrete carrier. We now give a single family in every active rank for which these layers can all be read off exactly. The example is deliberately uniform: the active rank tends to infinity, the physical permutation image remains the same finite group, the realized pure image remains nontrivial, and the buffered ancestry cost is constant, while the literal carrier height grows by an exact linear law.

9.1 Minimal fibers and a fixed finite physical image

For $s \geq 2$, let

$$P_s := \{0, e_1, \dots, e_s\} \subset \mathbb{Z}^s,$$

where $e_1, \dots, e_s$ is the standard basis.

Proposition 9.1 (Minimal marked fiber). *For every $s \geq 2$,*

$$\pi_1(Q(P_s)) \cong F_s = \langle x_1, \dots, x_s \rangle,$$

and under this identification the displacement map is the abelianization map

$$\delta_s : F_s \twoheadrightarrow \mathbb{Z}^s, \quad \delta_s(x_i) = e_i.$$

Consequently

$K_0(P_s) = [F_s, F_s], \quad K_{\text{lin}}(P_s) = 0.$

Proof. The points of P_s are affinely independent. Hence no two distinct nonzero pair differences agree, so the repeated-difference presentation has no nontrivial relation among the s generators corresponding to $e_1, \dots, e_s$. Thus

$$\pi_1(Q(P_s)) \cong F_s.$$

The displacement map sends the free basis to the standard basis of $\mathbb{Z}^s$, so its abelianization is an isomorphism

$$H_1(Q(P_s); \mathbb{Z}) \cong \mathbb{Z}^s \xrightarrow{\sim} \mathbb{Z}^s.$$

Therefore $K_{\text{lin}}(P_s) = 0$, and Corollary 3.4 gives $K_0(P_s) = [F_s, F_s]$. $\square$

Choose fixed permutations

$$r := (123), \quad q := (13)$$

in S_3 , and define

$$\rho_s : F_s \longrightarrow S_3$$

by

$$\rho_s(x_1) = r, \quad \rho_s(x_2) = q, \quad \rho_s(x_i) = 1 \quad (i \geq 3).$$

Since r and q generate S_3 , the image of ρ_s is independent of s .

Theorem 9.2 (Constant finite physical image in every rank). *For every $s \geq 2$,*

$$\boxed{\text{im } \rho_s = S_3, \quad \Pi_{\rho_s}(P_s) = A_3.}$$

In particular the physical image has constant order six and support three, while the pure physical image is nontrivial of order three for every active rank s .

Proof. The permutations $r = (123)$ and $q = (13)$ generate S_3 , so $\text{im } \rho_s = S_3$. By Proposition 9.1,

$$K_0(P_s) = [F_s, F_s].$$

For any surjective homomorphism $\varphi : G \twoheadrightarrow H$, one has

$$\varphi([G, G]) = [H, H].$$

Therefore

$$\Pi_{\rho_s}(P_s) = \rho_s([F_s, F_s]) = [S_3, S_3] = A_3.$$

Equivalently, already the commutator $[x_1, x_2]$ maps to the nontrivial 3-cycle $[r, q]$. Since the natural action of S_3 uses all three letters, the permutation support has cardinality three. $\square$

Thus active rank can grow arbitrarily while the finite physical transport complexity remains constant at the permutation level. The next step is to ask how much literal stack space is needed to realize the six persistent sheets of this fixed image.

9.2 A scalar counting realization

Consider the scalar weight vector

$$\omega_s = (\underbrace{1, \dots, 1}_{s+1 \text{ entries}}, -(s+1)).$$

At residual -1 , the minimum completions are the first $s+1$ unit vectors, so after normalization the optimal fiber is affinely equivalent to P_s . The all-ones vector lies in the integer kernel of ω_s ; it is a strictly positive zero-weight vector and will serve as the carrier seed below.

Lemma 9.3 (Exact six-sheet carrier girth). *For every $s \geq 2$,*

$$\boxed{g_6^{\text{ord}}(\omega_s) = s + 2.}$$

Proof. Let

$$z = (z_1, \dots, z_{s+2}) \in \mathbb{N}^{s+2}$$

be a nonzero zero-weight count vector. The equation $\omega_s z = 0$ is

$$\sum_{i=1}^{s+1} z_i = (s+1)z_{s+2}.$$

If $z_{s+2} = 0$, then every coordinate vanishes, contrary to $z \neq 0$. Hence $z_{s+2} \geq 1$, and

$$|z| = \sum_{i=1}^{s+1} z_i + z_{s+2} = (s+2)z_{s+2} \geq s+2.$$

This gives the lower bound

$$g_6^{\text{ord}}(\omega_s) \geq s+2.$$

For equality take

$$z = \mathbf{1} = (1, \dots, 1).$$

Then $\omega_s \mathbf{1} = 0$, $|\mathbf{1}| = s+2$, and

$$M(\mathbf{1}) = (s+2)! \geq 6.$$

Thus the vector $\mathbf{1}$ already supports at least six distinct literal orderings, proving

$$g_6^{\text{ord}}(\omega_s) = s+2.$$

□

9.3 Exact carrier height and ancestry cost

The all-ones kernel vector contains $s+2$ pairwise distinct literal types. Choose any three of them as an active top block. The six orderings of those three types are indexed by the elements of S_3 and serve as the regular persistent sheets. Keep the remaining $s-1$ types in a common fixed suffix. Thus every sheet has the same Parikh vector $\mathbf{1}$, zero semantic weight, and total height $s+2$.

Theorem 9.4 (All-rank exact buffered carrier family). *For every $s \geq 2$, the minimal active rank- s fiber admits a marked finite-permutation transport and buffered literal-order realization with*

$$\text{im } \rho_s = S_3, \quad \Pi_{\rho_s}(P_s) = A_3,$$

$$H_{\min} = s+2, \quad b_{\min} = 2.$$

Hence nontrivial pure physical holonomy arising from the nonabelian commutator sector persists in every active rank with constant finite permutation image and constant ancestry cost, while the exact literal carrier height grows linearly with slope one.

Proof. The physical-image statements are Theorem 9.2, and Proposition 6.2 supplies a flat local edge connection with this holonomy. Any nontrivial literal-order realization in the present category requires a nonempty zero-semantic carrier. Lemma 9.3 proves more than is needed for the lower bound: every nonzero $z \in \mathbb{N}^{s+2}$ with $\omega_s z = 0$ has $|z| \geq s+2$. Hence $H \geq s+2$ for every nontrivial persistent literal carrier in this scalar realization. The regular six-sheet construction above realizes the full S_3 transport with the all-ones Parikh vector at exactly height $s+2$, so the bound is attained.

The image S_3 acts on exactly three active carrier cells. Therefore

$$a(\text{im } \rho_s) = 3,$$

and Theorem 7.3 gives

$$b_{\min} = 3 - 1 = 2.$$

Thus both the persistent height and the buffered ancestry cost are exact. $\square$

Corollary 9.5 (Rank and physical complexity separate). *There is no lower bound forcing the order or support of the finite physical transport image to grow with active rank. In the family above, $s \rightarrow \infty$ while*

$$|\mathrm{im} \rho_s| = 6, \quad |\Pi_{\rho_s}(P_s)| = 3, \quad b_{\min} = 2$$

remain constant. At the same time the exact literal carrier height is $s+2$ because of the arithmetic zero-weight constraint of the chosen scalar realization.

Remark 9.6 (Real-time scope). Theorem 9.4 is exact in the buffered literal-order macro category fixed in Section 7. It does not assert that the same all-rank bounds are realizable in the stricter no- ε real-time Stable-LIFO subclass. That requires a separate scheduling construction and is not part of the present theorem package.

The family therefore isolates three distinct notions of size. Active semantic rank grows as s , finite holonomy complexity remains constant, and literal carrier height grows as $s+2$ for arithmetic rather than group-theoretic reasons. This is the concrete all-rank instance of the layered transfer principle developed above.

10 Discussion and Scope

The results above separate the completion problem into layers that are often merged in a single informal notion of “memory.” The intrinsic quotient first determines a marked group

$$\mathfrak{M}(P) = (G, L, \delta),$$

and therefore the available pure domain $K_0 = \ker \delta$ and its abelian shadow K_{lin} . A chosen representation ρ then decides which pure loops are physically visible. Only after that choice do literal carrier capacity and permutation support enter, and transient path memory and repair cost require still finer connection-level data. The resulting chain is

$\begin{aligned} \text{marked topology} &\rightarrow \text{physical holonomy image} \rightarrow \text{persistent carrier/locality} \\ &\rightarrow \text{transient connection memory} \rightarrow \text{repair cost.} \end{aligned}$
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The separation results above show that the layers studied here cannot, in general, be collapsed into a single scalar invariant.

Two rank-raising operations make this particularly transparent. Prism and apex both add one active lattice direction and both preserve K_{lin} . Nevertheless their full marked behavior is different. Prism forms a direct product and preserves K_0 , whereas apex forms a free product and adds zero-displacement commutators. The same distinction survives under finite physical transport: a prism extension preserves the realized pure image, while an apex can create new realized pure holonomy precisely when the new physical generator fails to centralize the old image. Thus active rank and first-homology visibility do not capture the nonabelian content of the transfer.

The higher-homology examples show a complementary limitation. The family $P_{m,s}$ has fixed marked group $(\mathbb{Z}^s, \mathbb{Z}^s, \mathrm{id})$ and therefore no pure zero-displacement sector at all, yet its higher Betti numbers grow without bound. Ordinary group-valued holonomy is a 1-type invariant: it is controlled by the fundamental group and, for flat edge transport, by relations in the 2-skeleton. Higher-dimensional homology can therefore vary independently. Detecting such higher topology

physically would require additional structure beyond the flat group-valued transport studied here; no such higher categorical transport is assumed in the present paper.

The resource statements are equally category-specific. The carrier-girth theorem assumes that permanent sheet identity is stored only in literal order at a common zero-semantic Parikh vector. The ancestry theorem assumes one-top ancestry-bijective buffered macros. The transient-state theorem applies to deferred-carrier connection certificates, and the repair coordinates are defined relative to a raw edge labeling, a chosen physical group, a cell presentation, and a metric. Changing any of these choices may change the numerical costs without changing $\mathfrak{M}(P)$. Accordingly, none of the results should be read as an unrestricted lower bound for every deterministic pushdown implementation.

The all-rank family makes these distinctions quantitative. Its semantic active rank is s , its full finite transport image remains S_3 , its pure image remains A_3 , and its exact buffered ancestry cost remains two. In contrast, the literal carrier height is $s + 2$. That growth is forced not by increasing permutation complexity but by the arithmetic zero-weight constraint of the scalar realization. Semantic dimension, finite-holonomy complexity, locality cost, and arithmetic carrier size are therefore independent coordinates even inside one explicit family.

Several extensions remain natural but are intentionally outside the present theorem package. The first is a stricter all-rank realization in the no- ε real-time Stable-LIFO subclass, where scheduling constraints may introduce additional transient state or locality costs. A second is the study of other physical image categories beyond finite permutations. A third is a genuinely higher transport theory capable of detecting information in H_2 and above rather than only in the marked fundamental group. These are refinements of the transfer layer, not gaps in the marked-group calculus proved here.

The conceptual conclusion is that higher-rank completion topology does transfer to physical pushdown structure, but only after the transfer category is specified. The displacement marking identifies the intrinsic zero-displacement domain; the chosen holonomy representation selects the physically visible part; and implementation-specific locality, connection, and repair data determine the actual resource cost. This layered formulation preserves the distinction between semantic normalization and physical realization while providing exact laws at each interface.

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